Amara Okonkwo

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Software Engineer transitioning into Cybersecurity, with firsthand experience in full-stack development, network security and secure software design. Secured vulnerable networks using industry best practices. Successfully developed and applied threat detection measures, security hardening techniques, and cryptographic solutions, applying frameworks like NIST, ISO 27001 and OWASP top 10. Now applying software engineering expertise to identify vulnerabilities, strengthen security controls, and protect organisations from cyber threats.

Skills

Cybersecurity Coding / Automation:	Python (security scripts, automation, scanning tools), Bash / PowerShell, Regex (log analysis, IDS rules).
Network & Infrastructure Security:	Firewalls, IDS/IPS (Snort, Suricata), Secure Network Protocols (TLS/SSL, IPsec, SSH, IKEv2, VPN).
Threat Detection, Monitoring & SIEM:	Splunk, Qualys.
Vulnerability Assessment & Penetration Testing:	Kali Linux, Nmap, Metasploit Framework, Wireshark, Gobuster.

Experience

Self — London
Student Cybersecurity Analyst — May 2025 - Present

- Training for CompTIA Security+ exam (Ongoing)
- Google Cybersecurity Certificate - Understand the importance of cybersecurity practices and their impact for organizations. Identify common risks, threats, and vulnerabilities, as well as techniques to mitigate them. Protect networks, devices, people, and data from unauthorized access and cyberattacks using Security Information and Event Management (SIEM) tools. Gain hands-on experience with Python, Linux, and SQL (Completed 2.11.2025)
- Qualys Vulnerability Management - Completed training in identifying, assessing, and managing security vulnerabilities across systems. Demonstrated understanding of vulnerability scanning, analysis, remediation workflows, and risk-based prioritization. Understood how to use the Qualys TruRisk platform for security and compliance. Covered asset visibility, vulnerability detection, risk scoring, and using Qualys tools to reduce exposure across an organisation. Configured and deployed Qualys sensors (virtual, cloud, network, or endpoint). Demonstrated foundational knowledge of how sensors collect security data and integrate into the broader Qualys ecosystem. (Completed 10.26.2025)

KPMG — Lagos, LOS
Intern Software Developer — November 2024 - April 2025

- Developed, integrated and tested a real-time voice-calling system using Agora SDK, OneSignal, Flutter, .NET, and Firebase—successfully deployed across KPMG internal tools.
- Reduced response latency by 25% by implementing a custom queue management system with load balancing and intelligent routing logic.
- Conducted full analysis and technical documentation of 5+ live production codebases (.NET, Node.js, React), improving onboarding efficiency for the dev team.
- Adapted rapidly to new technologies (Dart, Flutter), delivering production-ready features to a mobile app with 1,000+ active users within 3 weeks.
- Participated in client-facing meetings, live demos, and stakeholder updates—communicating technical progress and integration road maps.

IBM — Nottingham
Student Software Engineer — September 2022 - May 2023

- Delivered a formal proposal presentation to IBM, securing collaboration ahead of 56 competing teams in the university-wide competition..
- Developed interactive educational tools using Unity and C#, simulating key AI and software concepts for student use
- Leveraged IBM Cloud for backend storage and analytics, including IBM Watson for natural language insights on user interaction data.

Projects

Endpoint Detection & Response (EDR) Attack Simulation *LimaCharlie, Sliver C2, Windows Internals, Network Security*
https://github.com/Konkz7/CyberSecurityLabs/blob/main/SoyouwanttobeaSOCanalyst.pdf
Designed and executed a controlled attack-and-detect lab simulating a real-world SOC workflow. The project involved preparing a Windows 11 endpoint by disabling selected defensive controls, deploying a Sliver command-and-control payload from an Ubuntu attacker host, and collecting endpoint telemetry using LimaCharlie. Analysis focused on identifying suspicious process execution, correlating network activity to originating IP addresses, validating file hashes, and investigating timeline events for sensitive system access. The lab concluded with building custom Detection & Response rules to identify recurrence of the threat and alert on future malicious behaviour.

Wireshark Network Traffic Analysis *Wireshark, TCP/IP, Network Forensics*
https://github.com/Konkz7/Wireshark-Network-Analysis
Performed a full end-to-end packet capture and network investigation using Wireshark. The workflow included selecting the correct network interface, capturing live traffic, filtering by protocol (TCP, DNS, TLS), inspecting packet-level metadata, and analysing session flows for anomalies. The investigation concluded with exporting structured findings and building a formal technical report documenting the analysis process, key observations, and notable packet behaviours.

Metasploit Penetration Testing Workflow *Metasploit Framework, Vulnerability Exploitation*
https://github.com/Konkz7/Metasploit-Exploitation
Conducted a full exploitation cycle using the Metasploit Framework, including target enumeration, module selection, payload configuration, exploit execution, and post-exploitation operations to demonstrate real-world penetration testing methodology.

Hack The Box – Introductory Penetration Testing Labs *Linux, Windows, Enumeration, Exploitation*
https://github.com/Konkz7/HTB-StartingPoint
Completed the full Hack The Box *Starting Point* penetration-testing track, working through a range of beginner-to-intermediate machines. Each machine required applying structured methodologies including enumeration, service scanning, vulnerability discovery, credential extraction, privilege escalation, and post-exploitation analysis. Gained hands-on experience with diverse environments (Linux/Windows hosts) and attack vectors such as misconfigurations, weak credentials, exposed services, outdated software, and web-based flaws.

Education

University of Nottingham — London
BSc Hons Computer Science with Artificial Intelligence *Degree: 2:1* — Sep 2021 - July 2024

Mill Hill County High School — London
A-Level Computer Science / Chemistry / Maths *Degree(s): A A A* — Sep 2019 - July 2021

Hobbies

Football Has always been a lifelong passion of mine and I don't ever see that changing.

Brazilian Jiu - Jitsu I occasionally grapple and I love the sport. Competed for my university twice and managed to get on the podium.

Hiking and Scrambling In addition to casual traveling, I tend to try challenge myself while enjoying nature. This lead me to complete hikes like Scaffell Pike (highest point in England), Snowdon and Tryfan in Wales.